

Reduced Collatz dynamics is periodical and the period equals 2 to the power of the count of $x/2$

Wei Ren ^{a,b,c}

^aSchool of Computer Science, China University of Geosciences, Wuhan, China; ^bProvincial Key Laboratory of Multimodal Perceiving and Intelligent Systems, Jiaying University, Jiaying, China; ^cKey Laboratory of Computational Science and Application of Hainan Province, Haikou, China

ABSTRACT

In this paper, we prove that reduced dynamics on Collatz conjecture is periodical, and its period equals 2 to the power of the count of $x/2$ computation in the reduced dynamics. More specifically, if there exists reduced dynamics of x (that is, start from an integer x and the computation will go to an integer less than x), then there must also exist reduced dynamics of $x+P$ (that is, if starting from an integer $x+P$, then computation will go to an integer less than $x+P$), where P equals 2 to the power of L , and L is the total count of $x/2$ computations (i.e. computational times) in the reduced dynamics of x (note that, equivalently, L is also the length of the reduced dynamics of x). Therefore, the power (or usage) of this period property, which is discovered and proved in this paper, is—the study of the existence of reduced dynamics of x will result in the existence of reduced dynamics of $x+P$ (and iteratively $x+n^*P$, n is a positive integer). Hence, only the partition of integers needs to be verified for the existence of their reduced dynamics. Finally, if any starting integer x can be verified for the existence of its reduced dynamics, then Collatz conjecture will be True (due to our proposed Reduced Collatz conjecture).

ARTICLE HISTORY

Received 6 May 2024
Accepted 23 June 2025

KEYWORDS

Collatz conjecture; $3x+1$ problem; period; reduced dynamics

MATHEMATICS SUBJECT CLASSIFICATION

11Y55; 11B85; 11A07

1. Introduction and motivation

The Collatz conjecture can be stated simply as follows: Take any positive integer x . If x is even, divide it by 2 to get $x/2$. If x is odd, multiply it by 3 and add 1 to get $3x + 1$. Repeat the process again and again. The Collatz conjecture is that no matter what the integer (i.e. x) is taken, the process will always eventually reach 1.

The current known integers that have been verified are about 60 bits by T.O. Silva using normal personal computers (Oliveira & Silva, 1999, 2010). They verified all integers that are less than 60 bits. Ren et al., (2018) verified $2^{100000} - 1$ can return to 1 after 481,603 times of $3x + 1$ computation, and 863,323 times of $x/2$ computation, which is the largest integer being verified in the world. (Ren, 2019a) also pointed out a new approach for the possible proof of Collatz conjecture. (Ren, 2019b) proposed to use a tree-based graph to reveal two key inner properties in reduced Collatz dynamics: one is ratio of the count of $x/2$ over the count of $3x + 1$ (for any reduced Collatz dynamics, the count of $x/2$ over the count of $3x + 1$ is larger than $\ln 3 / \ln 2$), and the other is partition (all positive integers are partitioned regularly

corresponding to ongoing dynamics). Wei Ren et al. (Ren & Xiao, 2019) also proposed an automata method for fast computing Collatz dynamics. All source code and output data by computer programs in those related papers can be accessed in public repository (Ren, 2019c). Jeffrey P. Dumont et al. present beautiful graphs for the original Collatz dynamics (Dumont & Reiter, 2001, 2003) in 2003. The problem is original dynamics mix the characteristics of reduced dynamics and thus original dynamics presents certain chaos. Marc Chamberland reviews the works on Collatz conjecture (Chamberland, 2003) in 2006, and some aspects in available analysis results are surveyed. Jeffrey C. Lagarias edits a book on $3x + 1$ problem and also reviews the problem in 2010 (Lagarias, 2010).

Different from above paper, to the best of our knowledge, this is the first paper that proves the existence of the periodicity on reduced dynamics, and reveals the period with specific number—the length of reduced Collatz dynamics, e.g. if the reduced dynamics for x exists, then the reduced dynamics for $x + P$ exists and is the same as that of x , where P equals 2^L and L is the length of reduced dynamics of x in terms of I and O . Iteratively, the reduced

dynamics for $x + P + P \dots + P$ exists, and is the same as that of x . Hence, only the partition (e.g. partial) of integers needs to be verified for the existence of their reduced dynamics. Finally, if any starting integer x can be verified for the existence of its reduced dynamics, then Collatz conjecture will be True (due to our proposed Reduced Collatz conjecture).

2. Preliminaries

Notation 2.1

1. $[1]_2 = \{x | x \equiv 1 \pmod{2}, x \in \mathbb{Z}^+\}; [0]_2 = \{x | x \equiv 0 \pmod{2}, x \in \mathbb{Z}^+\}.$
2. $[i]_m = \{x | x \equiv i \pmod{m}, x \in \mathbb{Z}^+, m \geq 2, m \in \mathbb{Z}^+, 0 \leq i \leq m-1, i \in \mathbb{Z}^+ \cup \{0\}\}.$
3. $\min(S = \{\dots\})$: The minimal element in a set S .
4. $\max(S = \{\dots\})$: The maximal element in a set S .

In this paper, we only discuss positive integers (denoted as $\mathbb{Z}^+$). Any odd x will iterate to $3x + 1$, which is always even. Collatz computation afterward is always $x/2$. If combing these two as $(3x + 1)/2$, then the Collatz computation $T(x)$ can be defined as follows: $T(x) = (3x + 1)/2$ if x is odd; otherwise, $T(x) = x/2$. For the convenience in presentation, we denote $(3x + 1)/2$ as “ $I(x)$ ” (or just “ I ”) and $x/2$ as “ $O(x)$ ” (or just “ O ”). Indeed, “ I ” is named from “Increase” due to $(3x + 1)/2 > x$, and “ O ” is named from “dOwn” due to $x/2 < x$.

$T^{(k+1)}(T^{(k)}(x))$ ($k \in \mathbb{Z}^+$) means two successive Collatz computations, where $T^{(k+1)} = I$ if $T^{(k)}(x) \in [1]_2$, and $T^{(k+1)} = O$ if $T^{(k)}(x) \in [0]_2$. For simplicity by using less parentheses, we can rewrite it as $T^{(k)}T^{(k+1)}(x)$.

Iteratively, $T^{(k)}(T^{(k-1)}(\dots(T^{(1)}(x))))$ $k \geq 2, k \in \mathbb{Z}^+$ can be written as $T^{(1)} \dots T^{(k-1)}T^{(k)}(x)$, and $T^{(k)} = I$ if $T^{(1)} \dots T^{(k-1)}(x) \in [1]_2$ and $T^{(k)} = O$ if $T^{(1)} \dots T^{(k-1)}(x) \in [0]_2$.

Stopping time of $n \in \mathbb{Z}^+$ is defined as the minimal number of steps needed to iterate to 1:

$$s(n) = \inf\{k : T^{(1)} \dots T^{(k-1)}T^{(k)}(n) = 1\}.$$

$T(x)$ is either $(3x + 1)/2$ or $x/2$ (i.e. $T \in \{I, O\}$), the $s(n)$ is thus the count of $(3x + 1)/2$ computation plus the count of $x/2$ computation.

The Collatz computation sequence (i.e. original dynamics) of $n \in \mathbb{Z}^+$ is the sequence of Collatz computations that occurs from starting integer to 1:

$$d(n) = T^{(1)} \dots T^{(L-1)}T^{(L)},$$

where $T^{(1)} \dots T^{(L-1)}T^{(L)}(n) = 1$, $L = s(n)$, $T^{(i)} \in \{I, O\}, i = 1, L$.

For example, Collatz computation sequence from starting integer 3 to 1 is $IIOOO$, because $3 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. Thus, $s(3) = 5$. $d(3) = IIOOO$.

Definition 2.2. Function $IsMatched : x \times t \rightarrow b$. It takes as input $x \in \mathbb{Z}^+$ and $t \in \{I, O\}$, and outputs $b \in \{True, False\}$. If $x \in [1]_2$ and $t = I$, or if $x \in [0]_2$ and $t = O$, then output $b = True$; Otherwise, output $b = False$.

Simply speaking, this function checks whether the forthcoming Collatz transformation (i.e. $t \in \{I, O\}$) matches with the current transformed integer x .

Definition 2.3. Function $Substr : s \times i \times j$. It takes as input s, i, j , where $s \in \{I, O\}^{|s|}$, $1 \leq i \leq |s|$, $1 \leq j \leq |s| - (i - 1)$, and outputs s' where $s = s_a \| s' \| s_b$, $|s_a| = i - 1$, $|s'| = j$, $|s_b| = |s| - |s_a| - |s'|$ and “ $\|$ ” returns length.

Remark 2.4.

1. E.g., $Substr(IIOO, 1, 4) = IIOO$, $Substr(IIOO, 1, 3) = IIO$.
2. Especially, $Substr(s, 1, |s|) = s$. $Substr(s, |s|, 1)$ returns the last transformation in s . $Substr(s, 1, 1)$ returns the first transformation in s . $Substr(s, j, 1)$ returns the j -th transformation in s .
3. In other words, s' is a selected segment in s that starts from the location i and has the length of j .
4. Simply speaking, this function can obtain the Collatz transforms from i to $i + j - 1$ from a given inputting transform sequence (e.g. reduced dynamics) in terms of $s \in \{I, O\}^{|s|}$.
5. Note that, $Substr(\cdot)$ itself is a function. In other words, it can be looked as $Substr(\cdot)(\cdot)$. E.g. $Substr(IIOO, 1, 1)(3) = I(3) = (3 * 3 + 1)/2 = 5$, $Substr(IIOO, 1, 2)(3) = II(3) = I(I(3)) = I(5) = (3 * 5 + 1)/2 = 8$, $Substr(IIOO, 1, 3)(3) = IIO(3) = O(II(3)) = O(8) = 8/2 = 4$, $Substr(IIOO, 1, 4)(3) = IIOO(3) = O(IIO(3)) = O(4) = 4/2 = 2 < 3$.
6. It is worth to stress that, although in above definition $j \geq 1$, it can be extended to $j \geq 0$ by assuming $Substr(\cdot, \cdot, 0)(x) = x$.

Definition 2.5. Collatz conjecture.

$\forall x \in \mathbb{Z}^+, \exists L \in \mathbb{Z}^+$, such that $T^{(1)} \dots T^{(L-1)}T^{(L)}(x) = 1$ where $T^{(j)} \in \{I, O\}, j = 1, 2, \dots, L$. Recall that,

$IsMatch(Substr(T^{(1)} \dots T^{(L-1)}T^{(L)}, 1, i)(x), Substr(T^{(1)} \dots T^{(L-1)}T^{(L)}, i + 1, 1)) = True$, where $i = 0, 1, 2, \dots, L - 1$.

Obviously, Collatz conjecture holds when $x = 1$. In the following, we mainly concern $x \geq 2, x \in \mathbb{Z}^+$.

Definition 2.6. Reduced Collatz conjecture. $\forall x \in \mathbb{Z}^+, x \geq 2, \exists L \in \mathbb{Z}^+$, such that $T^{(1)} \dots T^{(L-1)} T^{(L)}(x) < x$ and $T^{(1)} T^{(i-1)} T^{(i)}(x) \not< x$, where $i = 1, \dots, L-1$. $T^{(j)} \in \{I, O\}$ for $j = 1, 2, \dots, L$. Recall that,
 $\text{IsMatch}(\text{Substr}(T^{(1)} \dots T^{(L-1)} T^{(L)}, 1, i)(x),$
 $\text{Substr}(T^{(1)} \dots T^{(L-1)} T^{(L)}, i+1, 1)) = \text{True}$, where
 $i = 0, 1, 2, \dots, L-1$.

The reduced Collatz computation sequence (i.e. reduced dynamics) of $n \in \mathbb{Z}^+$ is the sequence of Collatz computations that occurs from starting integer to 1:

$$d_r(n) = T^{(1)} \dots T^{(L-1)} T^{(L)},$$

where $T^{(1)} \dots T^{(L-1)} T^{(L)}(n) < n$, $T^{(1)} \dots T^{(i-1)} T^{(i)}(n) \not< n$, $i = 1, 2, \dots, L-1$, $T^{(j)} \in \{I, O\}, j = 1, L$.

For example, reduced Collatz computation sequence from starting integer 3 is *IIOO*, because $3 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 2$. Thus, $d_r(3) = \text{IIOO}$.

Theorem 2.7 Collatz conjecture is equivalent to Reduced Collatz conjecture.

The proof is straightforward (Ren, 2019b).

Obviously, reduced dynamics is more primitive than original dynamics, because original dynamics consists of reduced dynamics. Simply speaking, reduced dynamics are building blocks of original dynamics.

Remark 2.8.

1. Obviously, $d_r(x \in [0]_2) = O$.
2. $d_r(3) = \text{IIOO}$, $d_r(5) = \text{IO}$, $d_r(7) = \text{IIIOIOO}$, $d_r(9) = \text{IO}$, $d_r(11) = \text{IIOIO}$. Indeed, we design computer programs that output all $d_r(x)$, $\forall x \in [1, 99999999]$ (Ren, 2019a). From the data we discover the property—period and its relation to the length (or the total count of $x/2$ computation) in reduced dynamics—will be proved in the following part of this paper.
3. In fact, we proved some results on $d_r(x)$ for specific x , e.g. $d_r(x \in [1]_4) = \text{IO}$, $d_r(x \in [3]_{16}) = \text{IIOO}$, $d_r(x \in [11]_{32}) = \text{IIOIO}$, and so on. (More reduced dynamics that can be directly given are discussed in Ren (2019c)).
4. *IIOO* can be denoted in short as $I^2 O^2$. *IIIOIOO* can be denoted in short as $I^3 O I O^2$. In other words, we denote $\underbrace{I \dots I}_n$ as I^n , and we denote $\underbrace{O \dots O}_n$ as O^n where $n \in \mathbb{Z}^+, n \geq 2$. We also assume $I^1 = I$, $O^1 = O$. $I^0 = O^0 = \emptyset$ means no transformation occurs.

5. In fact, we formally proved that the ratio exists in any reduced Collatz dynamics. That is, the count of $x/2$ over the count of $3x+1$ is larger than $\log_2 3$ (Ren, 2020). The ratio can also be observed and verified in our proposed tree-based graph (Ren, 2019c).

Example 2.9. If there exists $d_r(x)$ ($x \in \mathbb{Z}^+, x \geq 2$), then

1. (1) $s(x) < x$, where $s = d_r(x)$;
2. (2) $\text{Substr}(s, 1, i)(x) \not< x$, where $i = 1, 2, \dots, |s| - 1$;
3. (3) $\text{IsMatched}(\text{Substr}(s, 1, j-1)(x), \text{Substr}(s, j, 1)) = \text{True}$, where $j = 1, 2, \dots, |s|$.

Remark 2.10.

1. (1) $s(x)$ is the first transformed integer that is less than the starting integer.
2. (2) $\text{Substr}(s, 1, i)(x)$ ($i = 1, 2, \dots, |s| - 1$) are all intermediate transformed integers.
3. (3) When $j = 1$, we have $\text{Substr}(s, 1, j-1)(x) = \text{Substr}(s, 1, 0)(x) = x$. $\text{Substr}(s, j, 1)$ is the first transformation.
4. (4) If $\text{Substr}(s, 1, j-1)(x)$ ($j = 2, \dots, |s|$) is current transformed integer, then $\text{Substr}(s, j, 1)$ is the next intermediate Collatz transformation.

Proposition 2.11. Given $x \in \mathbb{Z}^+$, if there exists $d_r(x)$, then $d_r(x)$ ends by O .

Proof. Straightforward, it is due to $I(x) = (3x+1)/2 > x$.

3. Period theorem

We will formally prove $d_r(x + 2^L) = d_r(x)$, $L = |d_r(x)|$, $d_r(x) \in \{I, O\}^{\geq 1}$ in this section. Note that, interestingly, L is indeed the count of $x/2$ computations in reduced dynamics (as each I includes one $x/2$ and each $x/2$ has one $x/2$).

3.1. Observations

Definition 3.1. Period. $\min(\{P | d_r(x+P) = d_r(x), x, P \in \mathbb{Z}^+\})$ is called the period of x .

Remark 3.2.

1. $\forall x \in [0]_2, d_r(x+2) = d_r(x) = O$, period $P = 2^{|d_r(x)|} = 2^{|O|} = 2^1 = 2$;
2. $\forall x \in [1]_4, d_r(x+4) = d_r(x) = \text{IO}$, period $P = 2^{|d_r(x)|} = 2^{|IO|} = 2^2 = 4$.
3. We thus concentrate on $x \in [3]_4$ in the following.

For easily understanding, we point out two concerns in the forthcoming proof.

1. Obviously, $d_r(3 + 16) = IIOO = d_r(3)$. We observed that during the computing of $IIOO(3 + 16)$, intermediate transformed integers (i.e. $I(3 + 16)$, $II(3 + 16)$, $IIO(3 + 16)$) are odd or even, if and only if $I(3)$, $II(3)$, $IIO(3)$ are odd or even. Besides, 16 is the minimal integer to satisfy above requirements.
2. Formally speaking, the parity of $s(x + P)$ and $s(x)$ are always identical, where $s = \text{Substr}(d_r(x), 1, i)$, and $i = 1, 2, \dots, |d_r(x)| - 1$. Also, the parity of $x + P$ and x are identical. That is, the parity sequence during computing for the reduced dynamics of starting integer $x + P$ is identical with that of starting integer x , which results in the occurred Collatz transformations of both are exactly identical. Besides, P is the minimal integer to satisfy above requirements. This is one concern.
3. The other concern is to prove $s(x + P) < x + P$ and $s(x) < x$ where $s = d_r(x)$; Also, $s(x + P) \not< x + P$ and $s(x) \not< x$ where $s = \text{Substr}(d_r(x), 1, i)$, $i = 1, 2, \dots, |d_r(x)| - 1$.

A new notation $I'(\cdot)$ is introduced hereby to reveal the relations among $I(x + P)$, $I(x)$ and $I'(P)$. Let $I'(x) = (3x)/2$.

Example 3.3.

1. $I(3+16) = (3(3+16)+1)/2 = (3*3+1)/2 + 3*16/2 = I(3) + I'(16)$, $I(3) = (3*3+1)/2 = 5$, $I'(16) = 3*16/2 = 24 \in [0]_2$, $5 > 3$, $5 + 24 > 3 + 16$. Thus, next transformations for $3 + 16$ and 3 are both I.
2. $II(3 + 16) = I(I(3) + I'(16)) = (3 * (I(3) + I'(16)) + 1)/2 = (3I(3) + 1)/2 + 3I'(16)/2 = II(3) + I'I'(16)$, $II(3) = I(5) = (3*5+1)/2 = 8$, $I'I'(16) = I'(24) = 3*24/2 = 36 \in [0]_2$, $8 > 3$, $8 + 36 > (3 + 16)$. Thus, next transformations for $3 + 16$ and 3 are both O.
3. $IIO(3 + 16) = O(II(3) + I'I'(16)) = IIO(3) + I'I'IO(16)$, $IIO(3) = 8/2 = 4$, $I'I'IO(16) = 36/2 = 18 \in [0]_2$, $4 > 3$, $4 + 18 = 22 > (3 + 16)$. Thus, next transformations for $3 + 16$ and 3 are both O.
4. $IIOO(3 + 16) = O(IIO(3) + I'I'IO(16)) = IIOO(3) + I'I'IOO(16)$, $IIOO(3) = 4/2 = 2$, $I'I'IOO(16) = 18/2 = 9$, $2 < 3$, $2 + 9 = 11 < (3 + 16)$. Thus, reduced dynamics for $3 + 16$ and 3 both end.

Remark 3.4. In the above example we can observe that $I'(16)$, $I'I'(16)$, $I'I'IO(16)$ are always remained even.

Thus, they do not influence the resulting next Collatz transformation (i.e. “I” or “O”) during computing for the reduced dynamics of starting integer 3. Therefore, the parity of $s(x + 16)$ and $s(x)$ is always identical, where $x = 3$, $s = \text{Substr}(d_r(x), 1, i)$, and $i = 1, 2, \dots, |d_r(x)| - 1$.

For better presentation, we thus introduce two functions as follows:

Definition 3.5. Function $\text{IsEven} : x \rightarrow b$. It takes as input $x \in \mathbb{Z}^+$, and outputs $b \in \{\text{True}, \text{False}\}$, where $b = \text{True}$ if $x \in [0]_2$ and $b = \text{False}$ if $x \in [1]_2$.

Definition 3.6. Function $\text{Replace} : s \rightarrow s'$. It takes as input $s \in \{I, O\}^{\geq 1}$, and outputs $s' \in \{I', O\}^{\geq 1}$, where $\text{Substr}(s', i, 1) = I'$ if $\text{Substr}(s, i, 1) = I$, and $\text{Substr}(s', i, 1) = O$ if $\text{Substr}(s, i, 1) = O$, for $i = 1, 2, \dots, |s|$.

Remark 3.7.

1. Simply speaking, replacing all “I” in “s” respectively by “I’” will result in “s’”.
2. Obviously, $\forall s \in \{I, O\}^{\geq 1}$, $|s'| = |s|$ where $s' = \text{Replace}(s)$.
3. By using above introduced functions, we can restate the reason in Remark 3.4 as follows: $\text{Substr}(s', 1, i) \in [0]_2$, where $s' = \text{Replace}(s)$, $i = 1, 2, \dots, |s'| - 1$, thus the parity of $s(x + 16)$ and $s(x)$ are always identical, where $x = 3$, $s = \text{Substr}(d_r(x), 1, i)$, and $i = 1, 2, \dots, |d_r(x)| - 1$.

3.2. The proof of period theorem

Lemma 3.8. If $P \in [0]_2$, $x \in \mathbb{Z}^+$, then $\text{IsEven}(x + P) = \text{IsEven}(x)$.

Proof. Straightforward. Due to $P \in [0]_2$, if $x \in [1]_2$, then $x + P \in [1]_2$; If $x \in [0]_2$, then $x + P \in [0]_2$. Thus, $\text{IsEven}(x + P) = \text{IsEven}(x)$.

Remark 3.9. Above lemma states that if $P \in [0]_2$, the first Collatz transformation of $x + P$ is identical to that of x .

Lemma 3.10. $s(x + P) = s(x) + s'(P)$, where $s \in \{I, O\}$, $s' = \text{Replace}(s)$, $x \in \mathbb{Z}^+$, $P \in [0]_2$.

Proof. $\text{IsEven}(x + P) = \text{IsEven}(x)$ because $P \in [0]_2$, due to Lemma 3.8. Thus, the first Collatz transformation of $x + P$ and the first Collatz transformation of x are identical.

1. Suppose $x \in [1]_2$, so $s = I$. Thus, $s' = \text{Replace}(s) = I'$.

$$s(x + P) = I(x + P) = 3((x + P) + 1)/2 = (3x + 1)/2 + 3 * P/2 = I(x) + I'(P) = s(x) + s'(P).$$

2. Suppose $x \in [0]_2$, so $s = O$. Thus, $s' = \text{Replace}(s) = O$.

$$s(x + P) = O(x + P) = (x + P)/2 = x/2 + P/2 = O(x) + O(P) = s(x) + s'(P).$$

Summarizing (1) and (2), $s(x + P) = s(x) + s'(P)$.

Lemma 3.11. (Separation lemma.) Suppose $x \in \mathbb{Z}^+$, $s \in \{I, O\}^{\geq 2}$, $s' = \text{Replace}(s)$. If $\text{Substr}(s', 1, j)(P) \in [0]_2$, $j = 0, 1, 2, \dots, |s| - 1$, then

1. $\text{IsEven}(\text{Substr}(s, 1, j)(x + P)) = \text{IsEven}(\text{Substr}(s, 1, j)(x))$;
2. $\text{Substr}(s, 1, j + 1)(x + P) = \text{Substr}(s, 1, j + 1)(x) + \text{Substr}(s', 1, j + 1)(P)$.

Proof.

1. $j = 0$.
2. $\text{Substr}(s', 1, j)(P) \in [0]_2$.
 $\text{Substr}(s', 1, j)(P) = \text{Substr}(s', 1, 0)(P) = P$, thus $P \in [0]_2$. Thus, $\text{IsEven}(x + P) = \text{IsEven}(x)$ due to Lemma 3.8. Thus, the intermediate next Collatz transformation of $x + P$ and x are identical.
3. $\text{Substr}(s, 1, j + 1)(x + P) = \text{Substr}(s, 1, 1)(x + P) \quad \because j = 0$
 $= \text{Substr}(s, 1, 1)(x) + \text{Substr}(s', 1, 1)(P) \quad \because \text{Lemma 3.10}$
 $= \text{Substr}(s, 1, j + 1)(x) + \text{Substr}(s', 1, j + 1)(P)$.
4. $j = 1$.
5. Due to (1), $\text{Substr}(s, 1, 1)(x + P) = \text{Substr}(s, 1, 1)(x) + \text{Substr}(s', 1, 1)(P)$.
 Besides, $\text{Substr}(s', 1, 1)(P) \in [0]_2$. Thus, $\text{IsEven}(\text{Substr}(s, 1, 1)(x + P)) = \text{IsEven}(\text{Substr}(s, 1, 1)(x))$. Thus, the intermediate next Collatz transformation of $x + P$ and x are identical.
6. There exist two cases as follows:
7. If $\text{Substr}(s, 1, j + 1) = \text{Substr}(s, 1, j) \parallel I$, then
 $\text{Substr}(s, 1, j + 1)(x + P) = (\text{Substr}(s, 1, j) \parallel I)(x + P) = I(\text{Substr}(s, 1, j)(x + P)) = I(\text{Substr}(s, 1, j)(x) + \text{Substr}(s', 1, j)(P)) \quad \because (1.2)$
 $= (3(\text{Substr}(s, 1, j)(x) + \text{Substr}(s', 1, j)(P)) + 1)/2 = (3 * \text{Substr}(s, 1, j)(x) + 1)/2 + 3 * \text{Substr}(s', 1, j)(P)/2 = I(\text{Substr}(s, 1, j)(x)) + I'(\text{Substr}(s', 1, j)(P)) = (\text{Substr}(s, 1, j) \parallel I)(x) + (\text{Substr}(s', 1, j) \parallel I')(P) = \text{Substr}(s, 1, j + 1)(x) + \text{Substr}(s', 1, j + 1)(P)$.
8. If $\text{Substr}(s, 1, j + 1) = \text{Substr}(s, 1, j) \parallel O$, then
 $\text{Substr}(s, 1, j + 1)(x + P) = \text{Substr}(s, 1, j) \parallel O(x + P) = O(\text{Substr}(s, 1, j)(x + P))$

$$= O(\text{Substr}(s, 1, j)(x) + \text{Substr}(s', 1, j)(P)), \quad \because (1.2)$$

$$= (\text{Substr}(s, 1, j)(x) + \text{Substr}(s', 1, j)(P))/2 = O(\text{Substr}(s, 1, j)(x)) + O(\text{Substr}(s', 1, j)(P)) = (\text{Substr}(s, 1, j) \parallel O)(x) + (\text{Substr}(s', 1, j) \parallel O)(P)$$

$$= \text{Substr}(s, 1, j + 1)(x) + \text{Substr}(s', 1, j + 1)(P).$$

(Note that, here $j + 1 = 2$. Recall that “ $\parallel$ ” is concatenation.)

9. Similarly, $j = 2$.

$$\text{Due to (2), } \text{Substr}(s, 1, 2)(x + P) = \text{Substr}(s, 1, 2)(x) + \text{Substr}(s', 1, 2)(P).$$

Besides, $\text{Substr}(s', 1, 2)(P) \in [0]_2$. Thus,

$$\text{IsEven}(\text{Substr}(s, 1, j)(x + P)) = \text{IsEven}(\text{Substr}(s, 1, j)(x)).$$

Thus, the intermediate next Collatz transformation of $x + P$ and x are identical.

(Note that, here $j = 2$).

Again, we can prove the following similar to (2.2).

$$\text{Substr}(s, 1, j + 1)(x + P) = \text{Substr}(s, 1, j + 1)(x) + \text{Substr}(s', 1, j + 1)(P).$$

(Note that, here $j + 1 = 3$.)

10. Similarly, we can prove $j = 3, \dots, |s'| - 1$, respectively and especially in an order.

Remark 3.12.

1. Obviously, above conclusion can be extended to include $|s| = 1$ by Lemma 3.8 and Lemma 3.10.
2. Separation lemma states the sufficient condition (i.e. $\text{Substr}(s', 1, j)(P) \in [0]_2$, $j = 0, 1, 2, \dots, |s| - 1$) for guaranteeing that all intermediate parities of transformed integers $x + P$ are exactly identical to those of x . That is, condition (1) states that for each intermediate computing step, either oddity in two results is the same; condition (2) states that for each intermediate computing step, either computational value in two results is the same.
3. Separation lemma is general, as s could be either original dynamics or reduced dynamics of certain $x \in \mathbb{Z}^+$. That is the reason we give a special name to this lemma for emphasizing its importance.
4. In the above proof, we assume the number of transformations will be s . Note that, it does not influence the conclusion, as we can use condition “ $j = 0, 1, 2, \dots$ ” instead of “ $j = 0, 1, 2, \dots, |s| - 1$ ” to omit the assumption on the number of transformations.

We next explore how to compute $\text{Substr}(s', 1, j)(x)$.

Definition 3.13. Function $\text{CntI}(\cdot)$. $\text{CntI} : s \rightarrow n$. It takes as input $s \in \{I, O\}^{\geq 1}$, and outputs $n \in \mathbb{N}$ that is the count of I in s .

Example 3.14. $\text{CntI}(\text{IIIO}) = 2$, $\text{CntI}(\text{III}) = 3$. Obviously, the function name stems from “Count the number of I” in given inputting parameter (i.e. reduced dynamics).

Lemma 3.15. Suppose $s \in \{I, O\}^{\geq 1}$, $s' = \text{Replace}(s)$, $x \in Z^+$, we have

$$\text{Substr}(s', 1, j)(x) = \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} * x, \quad j = 1, 2, |s|.$$

Proof.

1. $|s| = 1$. Thus, $j = 1$.
2. If $s = I$, then $s' = \text{Replace}(s) = I'$.
 $\text{Substr}(s', 1, j)(x) = \text{Substr}(I', 1, 1)(x) = I'(x) = 3 * x/2 = 3^1 * x/2^1$
 $= 3^{\text{CntI}(I)} * x/2^{|I|} = 3^{\text{CntI}(\text{Substr}(s, 1, j))} / 2^j * x.$
3. If $s = O$, then $s' = \text{Replace}(s) = O$.
 $\text{Substr}(s', 1, j)(x) = O(x) = x/2 = 3^0 * x/2^1 = 3^{\text{CntI}(O)} * x/2^{|O|}$
 $= 3^{\text{CntI}(\text{Substr}(s, 1, j))} / 2^j * x.$
4. $|s| \geq 2$.
5. $j = 1$.
6. If $\text{Substr}(s, 1, 1) = I$, then
 $\text{Substr}(s', 1, j)(x) = I'(x) = 3 * x/2 = 3^1 * x/2^1 = 3^{\text{CntI}(I)} * x/2^{|I|} = 3^{\text{CntI}(\text{Substr}(s, 1, j))} / 2^j * x.$
7. If $\text{Substr}(s, 1, 1) = O$, then
 $\text{Substr}(s', 1, j)(x) = O(x) = x/2 = 3^0 * x/2^1 = 3^{\text{CntI}(O)} * x/2^{|O|}$
 $= 3^{\text{CntI}(\text{Substr}(s, 1, j))} / 2^j * x.$
8. Iteratively, for $j = 1, 2, \dots, |s'| - 1$ in an order (recall that $|s'| = |s|$).
9. If $\text{Substr}(s, 2, 1) = I$, then
 $\text{Substr}(s', 1, j+1)(x)$
 $= (\text{Substr}(s', 1, j) \| I')(x)$
 $= I'(\text{Substr}(s', 1, j)(x))$
 $= 3 * \text{Substr}(s', 1, j)(x) / 2$
 $= 3 * \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} * x/2 \quad \because (2.1) \text{ for } j = 1, (2.2) \text{ for } j = 2, |s'| - 1$
 $= \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))+1}}{2^{j+1}} * x$
 $= \frac{3^{\text{CntI}(\text{Substr}(s, 1, j+1))}}{2^{j+1}} * x. \quad \because \text{Substr}(s, 1, j+1) = \text{Substr}(s, 1, j) \| I$
10. If $\text{Substr}(s, 2, 1) = O$, then
 $\text{Substr}(s', 1, j+1)(x)$
 $= (\text{Substr}(s', 1, j) \| O)(x)$
 $= O(\text{Substr}(s', 1, j)(x))$
 $= \text{Substr}(s', 1, j)(x) / 2$
 $= \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} * x/2$
 $= \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^{j+1}} * x$
 $= \frac{3^{\text{CntI}(\text{Substr}(s, 1, j+1))}}{2^{j+1}} * x. \quad \because \text{Substr}(s, 1, j+1) = \text{Substr}(s, 1, j) \| O$

Remark 3.16.

1. Recall that $s' = \text{Substr}(s', 1, |s|)$ and $s = \text{Substr}(s, 1, |s|)$. Thus, when $j = |s|$, then $s'(x) = \text{Substr}(s', 1, |s|)(x) = \frac{3^{\text{CntI}(\text{Substr}(s, 1, |s|))}}{2^{|s|}} * x = \frac{3^{\text{CntI}(s)}}{2^{|s|}} * x.$
2. Indeed, $j = |\text{Substr}(s, 1, j)|$, thus the lemma can be restated as
 $\text{Substr}(s', 1, j)(x) = \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^{|\text{Substr}(s, 1, j)|}} * x, j = 1, 2, \dots, |s|.$

Lemma 3.17.

$$\min(\{P | \text{Substr}(s', 1, j)(P) \in [0]_2, j = 0, 1, 2, |s| - 1, s \in \{I, O\}^{\geq 2}, s' = \text{Replace}(s), P \in Z^+\}) = 2^{|s|}.$$

Proof.

1. $j = 0$, $\text{Substr}(s', 1, j)(P) \in [0]_2 \Leftrightarrow P \in [0]_2.$
2. $j = 1, 2, \dots, |s| - 1.$

$$\text{By Lemma 3.15, } \text{Substr}(s', 1, j)(x) = \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} * x.$$

x. Thus,

$$\text{Substr}(s', 1, j)(P) = \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} * P \in [0]_2, j = 1, 2, |s| - 1$$

$$\Leftrightarrow P/2^j \in [0]_2, j = 1, 2, |s| - 1$$

$$\Leftrightarrow P/2^{j+1} \in Z^+, j = 1, 2, |s| - 1$$

$$\Leftrightarrow P/2^{|s|} \in Z^+$$

$$\Leftrightarrow \min(P) = 2^{|s|}.$$

$$\text{By (1) and (2), } \min(P) = 2^{|s|}.$$

Notation 3.18. $\text{Set}_{rd} = \{s | x \in Z^+, \exists d_r(x), s = d_r(x) \in \{I, O\}^{\geq 1}\}.$

Simply speaking, Set_{rd} is a set of all reduced dynamics for those $x \in Z^+$ if there exists reduced dynamics of x . That is, $\forall x \in Z^+$, if there exists $d_r(x)$, then $d_r(x) = s$ will be included in Set_{rd} , which is a set of existing reduced dynamics.

Lemma 3.19. $s'(P) < P$, where $s \in \text{Set}_{rd}$, $s' = \text{Replace}(s)$, $P \in Z^+.$

Proof. By Lemma 3.15 and Remark 3.16 (1),

$$s'(P) = 3^{\text{CntI}(s)} / 2^{|s|} * P.$$

Due to Corollary 4.4, $\frac{3^{\text{CntI}(s)}}{2^{|s|}} < 1$. Thus, $3^{\text{CntI}(s)} / 2^{|s|} * P < P$.

Remark 3.20. Here we use Corollary 4.4 (Form Corollary) given in Appendix, which is formally proved by us in another paper (Ren, 2020).

Lemma 3.21. $\text{Substr}(s', 1, j)(P) > P, j = 1, 2, |s| - 1$, where $s \in \text{Set}_{rd}$, $s' = \text{Replace}(s)$, $P \in Z^+.$

Proof. By Lemma 3.15,

$$\text{Substr}(s', 1, j)(P) = \frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} * P.$$

Due to Corollary 4.5, $\frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} > 1$. Thus $\frac{3^{\text{CntI}(\text{Substr}(s, 1, j))}}{2^j} * P > P$.

Theorem 3.22 (Period theorem.)

If there exists $d_r(x) \in \{I, O\}^{\geq 1}$, then there exists $d_r(x + 2^L)$, and $d_r(x + 2^L) = d_r(x)$ where $L = |d_r(x)|$.

Proof.

1. Regarding two special cases:

$$d_r(x \in [0]_2) = O, d_r(x + 2^L) = d_r(x), L = |d_r(x)| = |O| = 1.$$

$$d_r(x \in [1]_4) = IO, d_r(x + 2^L) = d_r(x), L = |d_r(x)| = |IO| = 2.$$

2. Next, without loss of generality, suppose $d_r(x) = s, |s| \geq 3$. Let $P = 2^L = 2^{|d_r(x)|} = 2^{|s|}$, $s' = \text{Replace}(s)$.

Thus, $\text{Substr}(s', 1, j)(P) \in [0]_2, j = 0, 1, 2, \dots, |s| - 1$, and P is the minimal integer for this requirement by Lemma 3.17.

3. Regarding the ordered parity sequence of $x + P$ and x .

$P \in [0]_2$, thus the first transformation of $x + P$ and x is identical by Lemma 3.8.

$\text{Substr}(s, 1, j + 1)(x + P) = \text{Substr}(s, 1, j + 1)(x) + \text{Substr}(s', 1, j + 1)(P)$ where $s' = \text{Replace}(s)$, by Lemma 3.11 (i.e. *Separation Lemma*). The parity for all transformed integers for $x + P$ and P (except for the last one) is exactly identical due to $\text{Substr}(s', 1, j)(P) \in [0]_2, j = 0, 1, 2, \dots, |s| - 1$.

4. Regarding the comparison between transformed integers and starting integer.
5. $s(x) < x$ due to the definition of reduced dynamics of x (recall that $s = d_r(x)$).
6. $\text{Substr}(s, 1, j)(x) \not< x, j = 1, 2, \dots, |s| - 1$, due to the definition of reduced dynamics of x (recall that $s = d_r(x)$).
7. $s'(P) < P$ by Lemma 3.19.
8. $\text{Substr}(s', 1, j)(P) \not< P, j = 1, 2, \dots, |s'| - 1$, by Lemma 3.21.
9. $s(x + P) = s(x) + s'(P) < x + P$, by Lemma 3.11, (2.2.1) and (2.2.3).
10. $\text{Substr}(s, 1, j)(x + P) = \text{Substr}(s, 1, j)(x) + \text{Substr}(s', 1, j)(P) \not< x + P, j = 1, 2, |s| - 1$, due to (2.2.2) and (2.2.4).

Thus, only the last transformed integer $s(x + P)$ is less than the starting integer $x + P$, and the other transformed integers are not less than the starting integer $x + P$. Thus, $d_r(x + P) = s$.

Due to (2.1) and (2.2), $d_r(x + P) = d_r(x), P = 2^L, L = |s| \geq 3$.

Summarizing (1) and (2), $d_r(x + 2^L)$ exists and $d_r(x + 2^L) = d_r(x)$ where $L = |d_r(x)| \in \mathbb{Z}^+$.

Remark 3.23. Interestingly, the period equals 2 to the power of the count of $x/2$ in reduced dynamics, as the count of $x/2$ equals to the length of $d_r(x)$. The count of $x/2$ has two folders: one equals the count of “I” due to $(3 * x + 1)/2$; the other equals the count of “O” due to $x/2$. Obviously, the count of “I” add the count of “O” equals the length of $d_r(x)$.

Corollary 3.24. If reduced dynamics of $x \in \mathbb{Z}^+$ exists, denoted as $d_r(x) \in \{I, O\}^L, L \in \mathbb{Z}^+$, then reduced dynamics of $x + k * 2^L$ ($k \in \mathbb{Z}^+$) exists and is identical to that of x .

Corollary 3.25. $\forall s \in \text{Set}_{rd}, \|\{x | d_r(x) = s\}\| = \|\mathbb{Z}^+\| = \aleph_0$ where “ $\|\cdot\|$ ” returns the number of a set.

Proof. Let $2^{|s|} = P \in \mathbb{Z}^+, d_r(x + P) = s = d_r(x)$. Thus, $d_r(x + k * P) = s = d_r(x), k \in \mathbb{Z}^+$. A bijective mapping from $\{x | d_r(x) = s\}$ to $\mathbb{Z}^+$ can be created as follows: $x \leftrightarrow 1, x + k * P \leftrightarrow k + 1, k \in \mathbb{Z}^+$. Thus, $\|\{x | d_r(x) = s\}\| = \|\mathbb{Z}^+\| = \aleph_0$.

4. Conclusion

We proved the main theorem called Period Theorem (i.e. Theorem 3.22) after the preparation of some lemmas, and we also present some corollaries of the main theorem. The major contribution of the paper is that we not only discover the period in the reduced dynamics, but also give what the period is or how to compute the period. The period for the reduced dynamics of x (i.e. $d_r(x) \in \{I, O\}^L$) is 2^L where L is the length of the reduced dynamics of x consisting of computations either $(3x + 1)/2$ (denoted as I computation) or $x/2$ (denoted as O computation). That is, $L = |d_r(x)|$. In other words, L is also the counts of $x/2$ computations, as each I has one $x/2$ computation and each O has one $x/2$ computation. In short, $d_r(x + 2^L) = d_r(x)$ where $L = |d_r(x)|$. That is, if there exists reduced dynamics of x , then there exists reduced dynamics of $x + 2^L$ and these two reduced dynamics are the same, where L is the length of the reduced dynamics of x . Iteratively, there exists the reduced dynamics of $x + k * 2^L$ where $k \in \mathbb{Z}^+$, and all are the same as the reduced dynamics of x .

The power of the Period Theorem is that it can be applied directly for the proof of Reduced Collatz Conjecture, which is equivalent to Collatz Conjecture (the equivalence is proved in our paper, see reference (Ren, 2019b)). The reason is that we only need to check partial integers (that is extremely less than before) for verifying or proving Collatz conjecture. That is, if $x \in \mathbb{Z}^+ (x \geq 2)$ is verified for the existence of reduced dynamics (namely, $d_r(x)$, which consists of I and O), then all integers in this

residue class (i.e. $x + k * 2^L$, $L = |d_r(x)|$, $k \in \mathbb{Z}^+$) have also been verified (i.e. verified automatically). With short notation, if $d_r(x)$ exists or is verified, then $d_r(x + P)$ is also verified where $P = 2^L$, $L = |d_r(x)|$. Therefore, what is left (for future work) is to verify that reduced dynamics of partial integers exists (partial can be understood as residue class basis). In other words, if the reduced dynamics of x is computed or verified, then reduced dynamics for all integers in the residue class in terms of $[x]_P$, $P = 2^L$, $L = |d_r(x)|$ have been computed (or verified). We can call it as residue class basis $[x]_P$ is verified. If the union of all residue classes is positive integers, then all residue classes can cover all positive integers and without overlap. Or, if the union of all residue class basis is 1 ($[x]_P$ can be looked as the portion of positive integers as $1/P$), then all residue classes cover all positive integers and without overlap. If all residue classes can cover all positive integers, then Reduced Collatz Conjecture will be true, and Collatz conjecture will be true too.

Funding

This work was supported by the Key Laboratory of Computational Science and Application of Hainan Province [JSKX202302]; Provincial Key Laboratory of Multimodal Perceiving and Intelligent Systems [MPIS202416].

Acknowledgements

The research was financially supported by the Provincial Key Laboratory of Multimodal Perceiving and Intelligent Systems, Jiaying University (No. MPIS202416), and the Key Laboratory of Computational Science and Application of Hainan Province (No. JSKX202302).

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data (Source Code in C, Computer Program Outputs) used to support the findings of this study have been deposited in the (Oliveirae & Silva, 1999, 2010; Ren et al., 2018) repository. (1) Wei Ren, Exploring properties in reduced Collatz dynamics, IEEE Dataport, 2018. <http://dx.doi.org/10.21227/ge9h-8a77> (2018). (2) Wei Ren, Verifying whether extremely large integer guarantees Collatz conjecture (can return to 1 finally), IEEE Dataport, <http://dx.doi.org/10.21227/fs3z-vc10> (2018). (3) Wei Ren, Exploring the ratio between the count of $x/2$ and the count of $(3*x+1)/2$ in original dynamics for extremely large starting integers asymptotically, IEEE Dataport. <http://dx.doi.org/10.21227/rxx6-8322> (2018). (4) Wei Ren, Exploring the inverse mapping from a dynamics to a residue class—inputting a reduced dynamics or partial dynamics and outputting a residue class, IEEE Dataport. <http://dx.doi.org/10.21227/qmzw-gn71> (2018). (5) Wei Ren, Reduced Collatz Dynamics for Integers from 3 to 999,999, IEEE Dataport. <http://dx.doi.org/10.21227/hq8c-x340> (2018). (6) Wei Ren, Collatz Automata and Compute Residue Class from Reduced Dynamics by Formula, IEEE Dataport. <http://dx.doi.org/10.21227/7z84-ms87> (2018). The explanation of above data can be accessed by our scientific data paper (Ren, 2019a). Moreover, the update of the data can be accessed by (7) Wei Ren, Programs, Data, and Testing Results on Collatz Dynamics of Extremely Large Integers, Science Data Bank, doi:10.57760/sciencedb.07210 (2023).

doi.org/10.21227/hq8c-x340 (2018). (6) Wei Ren, Collatz Automata and Compute Residue Class from Reduced Dynamics by Formula, IEEE Dataport. <http://dx.doi.org/10.21227/7z84-ms87> (2018). The explanation of above data can be accessed by our scientific data paper (Ren, 2019a). Moreover, the update of the data can be accessed by (7) Wei Ren, Programs, Data, and Testing Results on Collatz Dynamics of Extremely Large Integers, Science Data Bank, doi:10.57760/sciencedb.07210 (2023).

Public interest statement

This paper reveals a very interesting fact that the reduced Collatz dynamics is periodical. Reduced Collatz dynamics means that the computational procedures in terms of $(3*x+1)/2$ and $x/2$ starting from an integer x and the computation will go to an integer less than x . We discover that if there exists reduced dynamics of x , then there must also exist reduced dynamics of $x+P$ (that is, if starting from an integer $x+P$, then computation will go to an integer less than $x+P$). Thus, P is the period. We proved that P equals 2 to the power of L , and L is the total count of $x/2$ computations (i.e. computational times) in the reduced dynamics of x . Equivalently, L is also the length of the reduced dynamics of x in terms of $(3*x+1)/2$ and $x/2$.

Notes on contributor



Wei Ren is a full professor at the School of Computer Science, China University of Geosciences, Wuhan, China, since 2013. He was with the Department of Electrical and Computer Engineering, Illinois Institute of Technology, USA, in 2007 and 2008. He was with School of Computer Science, University of Nevada Las Vegas, USA, in 2006 and 2007. He was with the Department of Computer Science,

Hong Kong University of Science and Technology, in 2004 and 2005. He obtained Ph.D. degree in Computer Science from School of Computer Science and Technology, Huazhong University of Science and Technology, China, in 2006. He published more than 200 refereed papers, 2 monographs, and 6 textbooks. He obtained 50 patents and 10 innovation awards. He is a Distinguished Member of China Computer Federation.

ORCID

Wei Ren  <http://orcid.org/0000-0001-8590-1737>

References

- Chamberland, M. (2003). An update on the $3x+1$ problem. *Bulletin de la Societat Catalana de Matematiques*, 18, 19–45. https://chamberland.math.grinnell.edu/papers/3x_survey_eng.pdf
- Dumont, J. P., & Reiter, C. A. (2001). Visualizing generalized $3x+1$ function dynamics. *Computers & Graphics*, 25(5), 883–898. [https://doi.org/10.1016/S0097-8493\(01\)00129-7](https://doi.org/10.1016/S0097-8493(01)00129-7)

- Dumont, J. P., & Reiter, C. A. (2003). *Real dynamics of a 3-power extension of the $3x+1$ Function*, dynamics of continuous, discrete and impulsive Systems. *Series A: Mathematical Analysis*, 10, 875–893. https://online.watsci.org/abstract_pdf/2003v10/v10n6a/pdfv10n6a/2.pdf
- Lagarias, J. C. (2010). The ultimate challenge: The $3x + 1$ problem. In C. L. Jeffrey (Ed.), *The $3x+1$ problem: An overview* (pp. 3–29). American Mathematical Society.
- Oliveira, T., & Silva, (1999). Maximum excursion and stopping time record-holders for the $3x+1$ problem: Computational results. *Mathematics of Computation*, 68(225), 371–384. <https://doi.org/10.1090/S0025-5718-99-01031-5>
- Oliveira, T., & Silva, E. (2010). Empirical verification of the $3x+1$ and related conjectures. In J. C. Lagarias (Ed.), *The ultimate challenge: The $3x+1$ problem* (pp. 189–207). American Mathematical Society.
- Ren, W. (2019a). *Reduced Collatz dynamics data reveals properties for the future proof of Collatz conjecture* (Vol. 4). <https://doi.org/10.3390/data4020089>
- Ren, W. (2019b, April). A New approach on proving Collatz conjecture. *Journal of Mathematics*, Wiley, ID, 6129836. <https://doi.org/10.1155/2019/6129836>
- Ren, W. (2019c, December). Ratio and partition are revealed in proposed graph on reduced Collatz dynamics. *Proc. of 2019 IEEE Intl Conf on Parallel & Distributed Processing with Applications, Big Data & Cloud Computing, Sustainable Computing & Communications, Social Computing & Networking (ISPA/BDCloud/SocialCom/SustainCom)* (pp. 501–510). Ximen, China. <https://doi.org/10.1109/ISPA-BDCloud-SustainCom-SocialCom48970.2019.00077>
- Ren, W. (2020). A reduced Collatz dynamics maps to a residue class, and its count of $x/2$ over count of 3^*x+1 is larger than $\ln 3/\ln 2$. *International Journal of Mathematics and Mathematical Sciences*, Hindawi, 2020, ArticleID 5946759. <https://doi.org/10.1155/2020/5946759>
- Ren, W., Simin, L., Xiao, R., & Wei, B. (2018, October). Collatz Conjecture for $2^{1000000} - 1$ is True - Algorithms for Verifying Extremely Large Numbers. 2018 *IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI)*, Guangzhou, China (pp. 411–416). <https://doi.org/10.1109/SmartWorld.2018.00099>
- Ren, W., & Xiao, R. (2019, August). How to fast verify Collatz conjecture by Automata. *Proc. of IEEE 21st International Conference on High Performance Computing and Communications; IEEE 17th International Conference on Smart City; IEEE 5th International Conference on Data Science and Systems (HPCC/SmartCity/DSS)* (pp. 2720–2729). Zhangjiajie, China. <https://doi.org/10.1109/HPCC/SmartCity/DSS.2019.00382>

Appendix

In our another paper (Ren, 2020), we proved the following *Form Corollary* that states the requirements on the count of “O” and the count of “I” in any reduced dynamics.

Definition 4.1. $\text{CntO} : s \rightarrow n$. It takes as input $s \in \{I, O\}^{\geq 1}$, and outputs $n \in \mathbb{N}^*$ that is the count of “O” in s .

e.g., $\text{CntO}(\text{IIOO}) = 2$, $\text{CntO}(\text{IO}) = 1$.

Corollary 4.2. (Form Corollary.) $\forall s \in \{I, O\}^{\geq 1}$, $s \in \text{Set}_{\text{RD}}$, if and only if

(1) $s = O$; Or,

(2) $\text{CntO}(s) = \lceil \log_2 1.5 * \text{CntI}(s) \rceil$ and $\text{CntO}(s') < \lceil \log_2 1.5 * \text{CntI}(s') \rceil$ where $s' = \text{Substr}(s, 1, i)$, $i = 1, 2, \dots, |s| - 1$, $|s| \geq 2$.

We can denote $\log_2 1.5 = \lambda$. Roughly speaking, the count of “O” over the count of “I” is large enough when and only when the last transformed integer reaches. More specifically, once the count of $x/2$ over count of $3 * x + 1$ is larger than $\ln 3 / \ln 2$, reduced dynamics will be obtained (i.e. computed transformed integer will be less than starting integer).

Following conclusions are all derived from above *Form Corollary*.

Following corollary states that Set_{rd} can be constructed by generating $s \in \{I, O\}^{\geq 1}$ that satisfies aforementioned requirements instead of by conducting concrete Collatz transformations for all $x \in \mathbb{Z}^+$.

Corollary 4.3. $\text{Set}_{\text{rd}} = \{O\} \cup \{s | s \in \{I, O\}^L, L \in \mathbb{Z}^+, L \geq 2, \text{CntO}(s) = \lceil \log_2 1.5 * \text{CntI}(s) \rceil, \text{CntO}(s') < \lceil \log_2 1.5 * \text{CntI}(s') \rceil, s' = \text{Substr}(s, 1, i), i = 1, 2, L - 1\}$.

Proof. It is straightforward due to [Corollary 4.2](#).

Following corollary states the relations between $\text{CntI}(s)$ and $\text{CntO}(s) + \text{CntI}(s) = |s|$, $s \in \text{Set}_{\text{rd}}$. Note that, $\text{CntI}(s)$ is indeed equal to the count of $3x + 1$ computation, and $|s|$ is indeed equal to the total count of $x/2$ computation in reduced dynamics.

Corollary 4.4. $s \in \{I, O\}^{\geq 1}$, $s \in \text{Set}_{\text{rd}}$, we have

(1) $|s| \geq \lceil \log_2 3 * \text{CntI}(s) \rceil$; (2) $3^{\text{CntI}(s)} < 2^{|s|}$.

Proof.

1. When $s = O$, $\text{CntI}(s) = 0$, $|s| = 1$. $1 > \lceil \log_2 3 * 0 \rceil = 0$ by [Corollary 4.3](#).

When $s \neq O$, $s \in \text{Set}_{\text{rd}} \Rightarrow |s| = \lceil \log_2 1.5 * \text{CntI}(s) \rceil + \text{CntI}(s) = \lceil \log_2 1.5 * \text{CntI}(s) + \text{CntI}(s) \rceil = \lceil \log_2 3 * \text{CntI}(s) \rceil$ by [Corollary 4.3](#).

In summary, $s \in \text{Set}_{\text{rd}} \Rightarrow |s| \geq \lceil \log_2 3 * \text{CntI}(s) \rceil$, and note that “ $>$ ” is obtained when and only when $s = O$.

2. $|s| \geq \lceil \log_2 3 * \text{CntI}(s) \rceil \Rightarrow |s| \geq \log_2 3^{\text{CntI}(s)} \Rightarrow 3^{\text{CntI}(s)} \leq 2^{|s|} \Rightarrow 3^{\text{CntI}(s)} < 2^{|s|}$.

Corollary 4.5. $s \in \text{Set}_{\text{rd}}$, $s \in \{I, O\}^{\geq 2}$, we have $3^{\text{CntI}(\text{Substr}(s, 1, j))} > 2^j$, $j = 1, 2, \dots, |s| - 1$.

Proof. Let $s' = \text{Substr}(s, 1, j)$, $j = 1, 2, |s| - 1$. Obviously, $|s'| = j$.

$s \in \text{Set}_{\text{rd}} \Rightarrow \text{CntO}(s') < \lceil \log_2 1.5 * \text{CntI}(s') \rceil \quad \because \text{Corollary 4.2}$
 $\Rightarrow \text{CntO}(s') < \log_2 1.5 * \text{CntI}(s') \quad \because \log_2 1.5 \notin \mathbb{Q}, \text{CntI}(s), \text{CntO}(s) \in \mathbb{Z}^+$
 $\Rightarrow \text{CntO}(s') + \text{CntI}(s') < \log_2 3 * \text{CntI}(s')$
 $\Rightarrow |s'| < \log_2 3 * \text{CntI}(s') \Rightarrow 2^{|s'|} < 3^{\text{CntI}(s')}$
 $\Rightarrow 3^{\text{CntI}(\text{Substr}(s, 1, j))} > 2^j$.

Remark 4.6. By *Form Corollary* ([Corollary 4.2](#)), $\text{CntO}(s)$ can be computed from $\text{CntI}(s)$ directly for any reduced dynamics as follows:

1. (1) $s \in \{I, O\}^{\geq 1}$, $s \in \text{Set}_{\text{rd}}$, $\text{CntI}(s) = 1 \Rightarrow \text{CntO}(s) = \lceil \lambda * \text{CntI}(s) \rceil = \lceil \lambda * 1 \rceil = \lceil \log_2 1.5 * 1 \rceil = \lceil 0.58496250 * 1 \rceil = 1$.
2. (2) $s \in \{I, O\}^{\geq 1}$, $s \in \text{Set}_{\text{rd}}$, $\text{CntI}(s) = 2 \Rightarrow \text{CntO}(s) = \lceil \lambda * \text{CntI}(s) \rceil = \lceil \lambda * 2 \rceil = \lceil \log_2 1.5 * 2 \rceil = \lceil 0.58496250 * 2 \rceil = 2$.
3. (3) $s \in \{I, O\}^{\geq 1}$, $s \in \text{Set}_{\text{rd}}$, $\text{CntI}(s) = 3 \Rightarrow \text{CntO}(s) = \lceil \lambda * \text{CntI}(s) \rceil = \lceil \lambda * 3 \rceil = \lceil \log_2 1.5 * 3 \rceil = \lceil 0.58496250 * 3 \rceil = 2$.
4. (4) $s \in \{I, O\}^{\geq 1}$, $s \in \text{Set}_{\text{rd}}$, $\text{CntI}(s) = 4 \Rightarrow \text{CntO}(s) = \lceil \lambda * \text{CntI}(s) \rceil = \lceil \lambda * 4 \rceil = \lceil \log_2 1.5 * 4 \rceil = \lceil 0.58496250 * 4 \rceil = 3$.

Copyright of Research in Mathematics is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.